APERIO | AUTONOMOUS EARTH INTELLIGENCE PLATFORM

AUTHORITY DISASTER INTELLIGENCE ASSESSMENT • DEFORESTATION

Location: Forest Region, Assam, India • Generated: 07 Oct 2026 08:49 UTC • Status: APPROVED BY OPERATOR

Target Event	Deforestation • Forest Region	**Risk Probability**	**42% (HIGH)**
Affected Area	2.1 km²	**Model Confidence**	76% Indicative
Exposed Population	420	**Antecedent Rainfall**	72 mm (72h: 572mm)
Terrain Slope	11° (Steep escarpment)	**SAM Scar Boundary**	Validated (7.03 km²)
Satellite Observation	Sentinel-1 SAR + Sentinel-2 MSI	**Human Verification**	APPROVED BY OPERATOR

Multi-Agent Intelligence & Evidence Progression

[SYNTHETIC DEMO DATA NOTICE] This report record is derived from synthetic demonstration data (*aperio_synthetic_deforestation_dataset.csv*) and does not represent verified satellite ground truth.
1. Satellite Optical Ingestion: Simulated Sentinel-2 multi-spectral canopy reflectance with NDVI change calculation.
2. SAM Segmentation: Simulated Zero-Shot Segment Anything canopy loss mask.
3. XGBoost & Bayesian Sequential Update: Optical disturbance and SAR backscatter features updated beliefs to calibrated probability rating of **42%**.

Conservation Advisory & Ground Inspection Planning

Designation	Unit / Command	Distance / Status	Exposure
State Forest Division Monitoring Post (Assam)	Forestry Range Unit	1.5 km • Patrol Track Open	**MONITORING**
Forest Region Buffer Perimeter Station	Conservation Squad	3.2 km • Operational	**SURVEILLANCE**

Accountable Operator Review & Audit Trail

Reviewing Desk: APERIO Response Operations Desk
Operator Notes: Official authority report generated by Mission Control Operator
Operational Boundary: Decision Support Output only. All evacuations require local ground confirmation.